

A READ/WRITE RFID TAG FOR LOW COST APPLICATIONS

Rupert Howes, Andy Williams, Mick Evans¹

Introduction

There are many applications opening up in radio frequency identification (RFID) where tracking of a large number of different items, or counting of a large number of identical items is required. Often such items are low cost, requiring the tagging system to be of a low cost in comparison. Typically, these tags will be written to only a small number of times, even only once, and then disposed of.

This paper describes the design of a RFID tag targeted towards such low-cost applications. The tag is compatible with the SuperTAG™ protocol.

SuperTAG™ Protocol Background

Several RFID tag protocols are already in existence, popular among which are those based on a fast binary search anti-collision method, whereby the reader/interrogator communicates with a single tag, which has been assigned a unique identification number (ID), by assembling a list of the IDs present in the field. Each tag can subsequently be polled. Such a system requires a unique ID code for every tag. In situations where the field consists of a large number of identical items, and the reader is only interested in counting the number of items present, this may appear to be an inefficient use of what is a finite resource (ID numbers).

An alternative scheme is the SuperTAG™ protocol, which allows a single ID number to be assigned to all items of the same type. Typically a tag will transmit its code to the reader at random time intervals. Communication with a single tag is achieved by quietening-down all other tags in the field. This is achieved by the reader sending out an ACK signal to a transponder after receiving its code. The tag then either reduces the rate at which it transmits its code or switches off entirely. This switch off facility enables the counting of many identical tags. In Fast SuperTAG™ mode communication with a single tag is achieved by sending a MUTE signal to all tags in the field. The tag which has just started its transmission continues. All other tags stay muted until either a timeout or until an ACK signal is sent to the transmitting tag.

In the SuperTAG™ protocol, the reader communicates with the tag by inserting short gaps in its transmission. The timing of the gap determines the command being sent.

¹ The authors are with Matrics Ltd, Stroud, Gloucestershire, GL5 5NN

Design Targets

The tag is targeted towards low-cost, high volume applications where the device is written to only a few times and the tag's lifetime is relatively short. Such applications might include retailing, stock control, parcel freight, airline baggage etc. where the cost of the tag needs to be reduced to a minimum.

As with other chip-based RFID tags, a key system design goal is to include all the required components on a single IC for cost reduction purposes. Because the main operating frequency for this tag is the 13.56MHz band, this necessitates an external coil antenna. This antenna is typically printed onto a thin, flexible, supporting film. The only additional item, then, is the IC itself.

A second key design goal is to minimize the power consumption to enable operation at maximum range. This requires special low voltage and low current design techniques and a suitable low power process.

A third goal is to minimize the chip area, again for minimal cost, requiring a small geometry process.

The process chosen is a 0.6 μ m, double polysilicon, double metal, embedded EEPROM process.

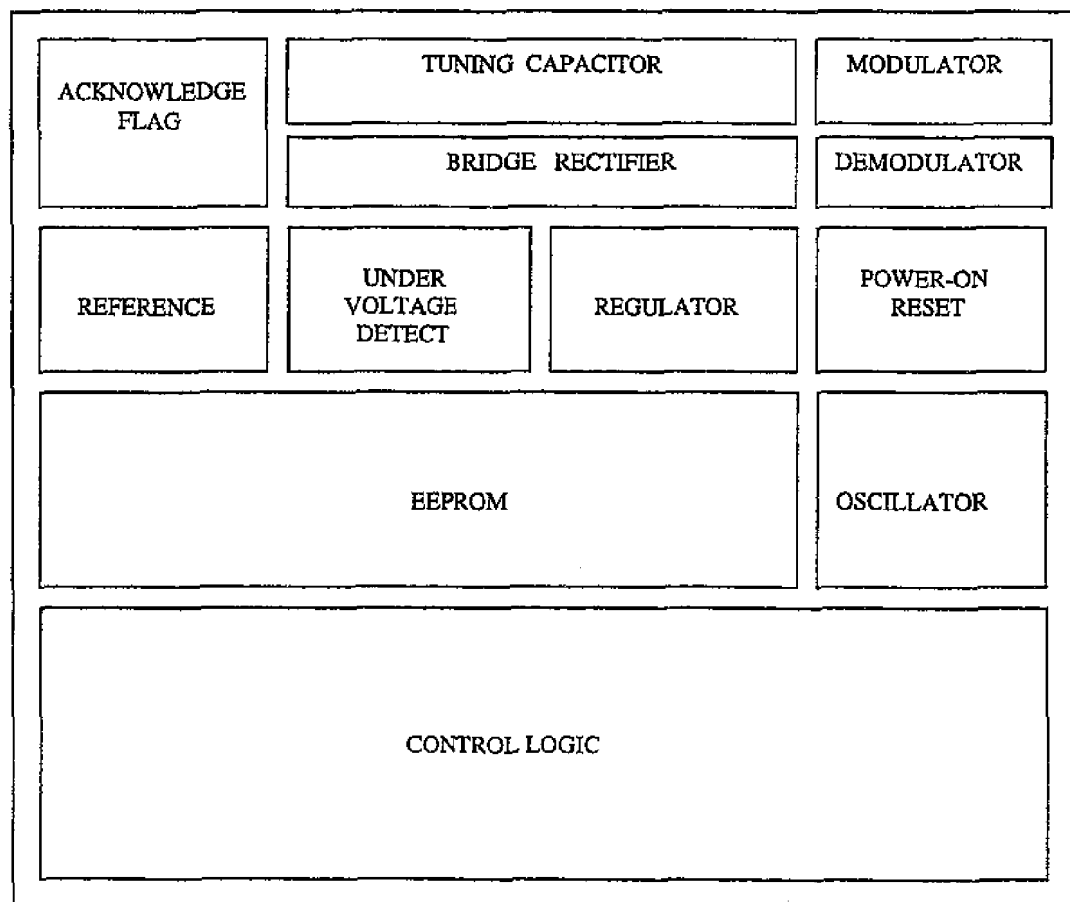


Figure 1

Design Features

The block diagram of the IC is shown in figure 1 and contains the following functional blocks :

1. Digital Control Logic to perform all the decoding and coding of the received and transmitted messages respectively. It also controls the writing and reading of data to and from the EEPROM.
2. EEPROM Memory to store configuration information and application specific data.
3. Oscillator to provide a system clock.
4. Modulator to signal from tag to reader.
5. Demodulator to receive messages from reader.
6. Reference to provide a voltage reference and bias currents for the various circuit blocks.
7. Power-on-Reset to initialise the circuit upon first power-up.
8. Under voltage detector to determine supply voltage level.
9. Voltage regulator to limit supply voltage.
10. Acknowledge flag to store 'acknowledge' condition.
11. Tuning capacitor.
12. Power storage capacitor.
13. Bridge rectifier for ac to dc conversion.

Circuit Operation

Power is supplied to the tag, in the case of 13.56MHz operation, through the two coil connections. This coil resonates with the on-chip tuning capacitor and the resulting ac signal is rectified by the bridge rectifier. The dc voltage is smoothed with a capacitor of approximately 2nF, which also provides the chip with power during carrier breaks.

The actual supply voltage is sensed by the under voltage detector to ensure it is sufficient for correct operation of the device. If the supply voltage is too low then the chip will maintain itself in a power down condition and ensures that the modulator is off. If the supply voltage becomes too high then the voltage regulator cuts in to prevent damage to the device. Upon initial power up the power-on-reset circuit ensures that the logic circuitry is reset to a known state.

Transmission from tag to reader and from reader to tag is by 100% carrier modulation, implemented by applying a temporary low impedance across the coil terminals. Encoding of tag to reader transmitted data is either Manchester or Glitch. Because of the 100% modulation the circuit cannot rely on deriving its system clock from the carrier. Instead an on-board RC oscillator provides the system clock. The frequency is independent of supply voltage, to first order, so it maintains a relatively tight tolerance over operating conditions.

The demodulator detects gaps in the carrier transmission, and converts these gaps to a logic signal. The decoding and interpretation of this data is performed by the digital circuitry.

The acknowledge flag block implements the SuperTAG™ ACK command by storing the logic state of the ACK for approximately 60 seconds in the absence of power.

The EEPROM is 1024 bits arranged in 8bit bytes and includes on-chip high voltage generation. Reading and writing is one byte at a time. All but 48 bits are available by the user application. These 48 bits are retained for configuration data such as switch-off/quiet down mode, encoding type

and data rate. Special attention has been given to the design of the EEPROM for low power operation both in read and write modes, to ensure maximum read and write ranges to/from the tag.

Figure 2 shows a more detailed block diagram for the control logic.

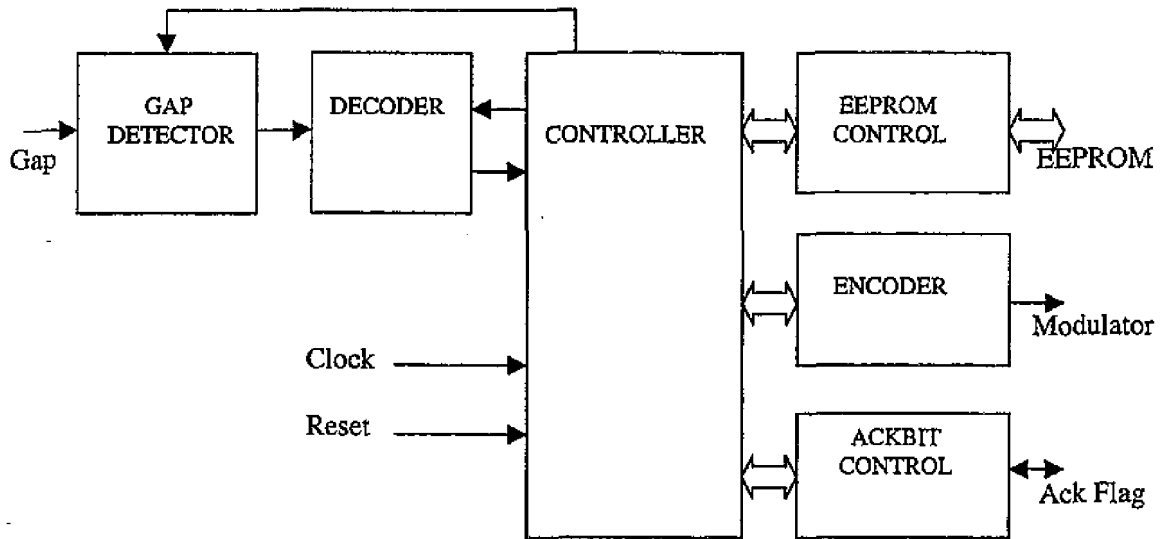


Figure 2

The gap detector detects the presence of intentional breaks in the carrier. The decoder extracts commands and data from the output of the gap detector. This information is coded in a pulse width modulated format where the time between gaps is decoded into logic 0's and 1's. The purpose of the encoder is to encode EEPROM data into either Manchester II or Glitch format data. The Ackbit control circuit allows setting and resetting of the Acknowledge flag. The EEPROM control generates the control signals for the read, write and erase operations of the EEPROM. The main controller is composed of several sub-units including the main state machine and protocol controller.

When initially powered up the logic will read the configuration bits from EEPROM and determine whether the device is a blank tag, i.e. it has yet to be initialised, or whether it is a coded tag. If a coded tag, then it will start to transmit its configuration data.

Summary

The design of a SuperTAG™ compatible RFID tag has been presented. This tag has been designed primarily for lost cost, high volume applications at 13.56MHz. It is implemented in a 0.6µm, double polysilicon, double metal, embedded EEPROM process occupying an area of 3.2mm².

Acknowledgements

BTG International Ltd for permission to publish this paper.
CSIR, South Africa for helpful suggestions and advice.

SuperTAG™ is a trademark of BTG International Ltd.